

Dynamic Host Configuration Protocol (DHCP)

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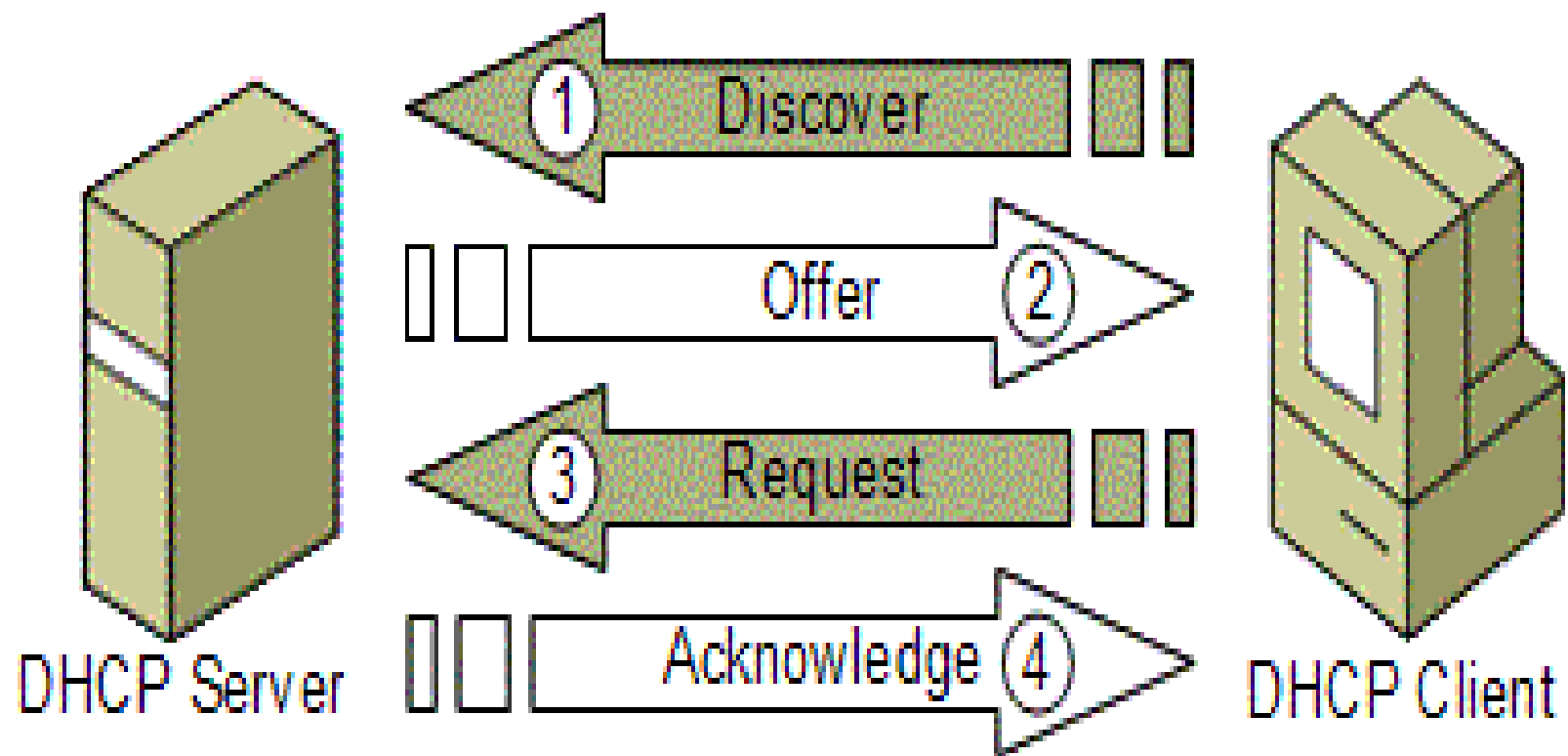
- Communication protocol that lets network administrators manage centrally and automate the assignment of Internet Protocol addresses in an organization's network. (basically it frees a network administrator from having to manually configure each host connected to his network)
- DHCP is an extension of the Bootstrap protocol (BOOTP).
- BOOTP was designed for manual pre-configuration of the host information in a server database.

Components of DHCP

- DHCP Server
- DHCP Client
- DHCP/BOOTP relay agent

DHCP Messages

- **DHCP DISCOVER** - Client broadcast to locate available servers.
- **DHCP OFFER** - Server to client in response to DHCP DISCOVER with offer of configuration parameters.
- **DHCP REQUEST** - Client message to servers either
 - (a) requesting offered parameters from one server and implicitly declining offers from all others,
 - (b) confirming correctness of previously allocated address after, e.g., system reboot, or
 - (c) extending the lease on a particular network address.
- **DHCP ACK** - Server to client with configuration parameters, including committed network address.



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802.11 Channel: Channel Offset: FCS Filter: Decryption Mode: Wireshark

No.	Len	Time	Source	Destination	Protocol	Info
1	314	0.000000	0.0.0.0	255.255.255.255	DHCP	DHCP Discover - Transaction ID 0
2	342	0.000295	192.168.0.1	192.168.0.10	DHCP	DHCP Offer - Transaction ID 0
3	314	0.070031	0.0.0.0	255.255.255.255	DHCP	DHCP Request - Transaction ID 0
4	342	0.070345	192.168.0.1	192.168.0.10	DHCP	DHCP ACK - Transaction ID 0

User Datagram Protocol, Src Port: bootpc (68), Dst Port: bootps (67)

Bootstrap Protocol

- Message type: Boot Request (1)
- Hardware type: Ethernet
- Hardware address length: 6
- Hops: 0
- Transaction ID: 0x00003d1d
- Seconds elapsed: 0
- Bootp flags: 0x0000 (Unicast)
- Client IP address: 0.0.0.0 (0.0.0.0)
- Your (client) IP address: 0.0.0.0 (0.0.0.0)
- Next server IP address: 0.0.0.0 (0.0.0.0)
- Relay agent IP address: 0.0.0.0 (0.0.0.0)
- Client MAC address: Grandstr_01:fc:42 (00:0b:82:01:fc:42)

DHCP Messages (..Contd)

DHCP NAK - Server to client indicating client's notion of network address is incorrect (e.g., client has moved to new subnet) or client's lease has expired

DHCP DECLINE - Client to server indicating network address is already in use.

DHCP RELEASE - Client to server relinquishing network address and canceling remaining lease.

DHCP INFORM - Client to server, asking only for local configuration parameters; client already has externally configured network address.

Allocating new network address

Step-by-step

- Client broadcasts DHCPDISCOVER
- Server respond with DHCPOFFER
- Server check for address
- Client broadcasts DHCPREQUEST
- Server selected commits binding, sends DHCPACK
- Server sends DHCPNAK / Client responds to DHCPACK
- Client sends DHCPDECLINE if address in use
- Client responds to DHCPNAK
- No DHCPACK/DHCPNAK
- Client sends DHCPRELEASE

Reusing previous network address

- Client broadcasts DHCPREQUEST
- Server responds with DHCPACK
- Server SHOULD NOT check for network address
- Client is configured
- Client detects address in use sends DHCPDECLINE
- Client receives DHCPNAK
- No DHCPACK/DHCPNAK
- Client sends DHCPRELEASE